

Yafira Martinez

Design Engineer / Creative Technologist

interactive systems, interfaces, and experimental technologies

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EXPERIENCE

Teaching and Academic Support

Graduate Assistant — Intro to Wearables, NYU (IMA)

Jan 2026 – Present

Instructor: Daniel Ryan Johnston

- Support instruction in wearable and e-textile design, focusing on body-centered interaction, gesture, and non-verbal communication.
- Assist students with hands-on projects with soft circuits and Adafruit Gemma M0, using Arduino code with light and motion sensors to prototype expressive wearables.

Graduate Assistant — Creative Computing, NYU (IMA)

Sept 2025 – Dec 2025

Instructor: Daniel Shiffman

- Supported Daniel Shiffman in teaching and mentoring students in creative coding, including office hours, class workshops, and assignment feedback.

Research Assistant — Computational Text, NYU (ITP/IMA)

May 2025–Aug 2025

Advisor: Daniel Shiffman

- Assisted in developing the Computational Text course syllabus by identifying tools, artist projects, and frameworks in procedural text, data ethics, and machine learning.
- Researched and curated inclusive readings, case studies, and interactive works that broaden creative coding through experimental approaches.

Research & Open Source

Machine Learning Researcher & Web Contributor — ml5.js

June 2025 – Present

- Develop ml5.transcribe() and ml5.embeddings(), expanding access to browser-based transformer models (Whisper, Moonshine) for creative coding.
- Refine interaction design, inference performance, and web tooling to support real-time creative ML and contributor usability.

Accessibility Researcher — Making Fashion Accessible, Vertically Integrated Projects (VIP), NYU Ability Lab

Oct 2025–Present

- Research accessible fashion technologies using knitting machines and AYAB to support inclusive, open-source garment production.
- Contribute to adaptable textile workflows centered on accessibility and human-centered design.

Development & Community

Lead Web Developer — Equitable Syllabus Project, NYU (ITP/IMA)

Nov 2024 – Present

- Design and implement accessible, user-centered interfaces using React.js and Figma.
- Integrate and migrate project data while maintaining centralized contribution tracking (Google Sheets) to improve usability and sustainability.

Web Developer & Designer — Useless Machines Archive

Mar 2025 – Aug 2025

Instructor: Blair Simmons

- Designed and built a front-end web archive using the Are.na API, focusing on client-side data fetching, content organization, and responsive interface design.

Camp Counselor — ITP Camp, New York University

June 2025 - July 2025

- Provided technical, operational, and community support for 150+ participants across creative technology workshops and events.
- Managed help desk troubleshooting, space setup, internal communications, and final showcase coordination.

SKILLS

Front-End/Web Development

UX/UI Design

Interaction Design

Creative Coding

Data Analysis

Technical Writing & Documentation

Basic 3D Modeling

3D Printing

Physical Computing

TOOLS

Visual Studio Code, Figma, Git, Github,

Arduino IDE, Fritzing, Adobe Suite,

Obsidian, Notion, WordPress

CERTIFICATIONS

Artificial Intelligence,

Samsung Innovation Campus

November 2022 – May 2023

Human Computer Interaction,

Interaction Design Foundation

February 2022

Google UX Design, Coursera

July 2021 – December 2021

EDUCATION

Master of Professional Studies in

Interactive Telecommunications

Program (ITP)

New York University –

Tisch School of Arts

Expected: May 2026

Bachelor of Science, Computer Science

City University of New York -

Lehman College

December 2020

INTERESTS

Soft computing and interfaces; electronics and e-textiles; wearable technologies; inclusive and open-source tools; experimental interaction design; creative research; electronic text, digital archives, & creative writing.

SELECTED WORK, TALKS & CREATIVE EXPLORATIONS

Pocket Oracle of Time — Horological Society of New York (HSNY)

Feb 2026 - present

- Exhibiting a poetic timekeeping object exploring ritual, slowness, and material interaction with time through embedded electronics and tactile design.

Moon Pocket — Pedestal Playtime: Art Toy & Tech Toy Open Mic, NYC Resistor

Jan 2026

- Showcased a connected device built with an Arduino Nano 33 IoT that fetches real-time moon and tide data from public web APIs and renders it on a round TFT display through touch-based interaction.

Tendertronics: Analog Circuit Performance — NYU ITP's New Interfaces for Musical Expression (NIME),

Littlefield

Dec 2025

- Performed an analog instrument exploring tactile, minimal interfaces and embodied sound interaction through handcrafted circuitry.

Lightning Talk Speaker — Exploring Creativity in the Age of AI, NYU Global Show & Tell

Oct 2025

- Presented *Open Tools, Soft Interfaces*, discussing contributions to ml5.js and open-source machine learning, and framing creativity in AI through openness, design as dialogue, and soft, human-centered interfaces.

UN Open Source Week Hackathon, United Nations Headquarters

June 2025

- Participated in a global hackathon focused on open-source solutions for child-centered climate emergency response; contributed to the Ahead of the Storm challenge using geospatial data.

The Internet Book — Annual publication on the poetic web

May 2025

- Featured for electrocute.io, a collection of soft-tech and poetic computing experiments, in a curated directory of designers, developers, writers, and digital artists.

Static Chic — Volt Couture: Wearables + E-textiles Open Mic, NYC Resistor

May 2025

- Presented *Static Chic*, a 3D-printed inductive handbag that wirelessly lights up embedded LEDs. Part of a wearables and e-textiles open mic curated by Kari Love and Liza Stark.

Rejects: A Benefit Auction, CultureHub

May 2025

- Contributed three oddities from my personal archive—*Gothic Teacrypt*, *Orbit Offtrack*, and *Pixel Shell*—to Rejects, a curated auction of discarded prototypes celebrating the charm of creative misfires.

Soft Glitch — eTextile Swatch, Radiant Fibers Exhibition, NYU Clive Davis Gallery

Jan 2025 - Mar 2025

- Exhibited an e-textile swatch combining punch needle, chaotic yarns, and electroluminescent thread to explore digital disruption through texture and light.

MIT Reality Hack 2025 — Massachusetts Institute of Technology

Jan 2025

- Implemented sound and haptic feedback for *HangarPro VR*, a virtual reality training simulation using Cognitive3D to enhance pre-flight safety through visual interaction analysis for aerospace exploration.

EcoThreads: Sustainable Smart Textiles — Hybrid Body Lab, Cornell Tech

Oct 2024

- Explored sustainable e-textile fabrication by creating biodegradable thermo-chromic conductive threads using a DIY wet-spinning process.